

variability within each category (i.e., more samples taken in more variable categories such as wetlands and fewer in less variable categories such as open water). As noted earlier, however, all decisions about sampling design must include consideration of the statistical validity and tractability of the accuracy assessment process.

Final Thoughts on Accuracy Assessment

There are three other facets of classification accuracy assessment that we wish to emphasize before leaving the subject. The first relates to the fact that the quality of any accuracy estimate is only as good as the information used to establish the “true” land cover types present in the test sites. To the extent possible, some estimate of the errors present in the reference data should be incorporated into the accuracy assessment process. It is not uncommon to have the accuracy of the reference data influenced by such factors as spatial misregistration, image interpretation errors, data entry errors, and changes in land cover between the date of the classified image and the date of the reference data. The second point to be made is that the accuracy assessment procedure must be designed to reflect the intended use of the classification. For example, a single pixel misclassified as “wetland” in the midst of a “corn” field might be of little significance in the development of a regional land use plan. However, this same error might be unacceptable if the classification forms the basis for land taxation or for enforcement of wetland preservation legislation. Finally, it should be noted that remotely sensed data are normally just a small subset of many possible forms of data resident in a GIS. The propagation of errors through the multiple layers of information in a GIS is beyond the scope of this work; interested readers may consult the literature on GIS and spatial analysis.

7.18 CHANGE DETECTION

One of the most powerful advantages of remote sensing images is their ability to capture and preserve a record of conditions at different points in time, to enable the identification and characterization of changes over time. This process is referred to as *change detection* and is among the most common uses of digital image analysis. The types of changes that might be of interest can range from nearly instantaneous (movement of a vehicle or animal) to long-term phenomena such as urban fringe development or desertification. Ideally, change detection procedures should involve data acquired by the same (or similar) sensor and be recorded using the same spatial resolution, viewing geometry, spectral bands, radiometric resolution, and time of day. For detecting changes at time scales greater than one year, *anniversary dates* are preferred to minimize sun angle and seasonal differences. Accurate spatial registration of the various dates of imagery

(ideally to within $\frac{1}{4}$ to $\frac{1}{2}$ pixel) is also a requirement for effective change detection. Clearly, when misregistration is greater than one pixel, numerous errors will result when comparing the images.

The reliability of the change detection process may also be strongly influenced by various environmental factors that might change between image dates. In addition to atmospheric effects, such factors as water level, tidal stage, wind, or soil moisture condition might also be important. Even with the use of anniversary dates of imagery, such influences as different planting dates and season-to-season changes in plant phenology must be considered.

One way of discriminating changes over time is to employ *postclassification comparison*. In this approach, two dates of imagery are independently classified and registered, and pixels whose class changed between dates are identified. In addition, statistics (and change maps) can be compiled to express the specific nature of the changes between the dates of imagery, for example, by highlighting only those areas that changed from *Class A* to *Class B*. Obviously, the accuracy of such procedures depends upon the accuracy of each of the independent classifications used in the analysis. The errors present in each of the initial classifications are compounded in the change detection process.

Figure 7.55 shows a representative example of postclassification change detection in a rapidly changing landscape. The original classifications for two dates appear in (a) and (b). In Figure 7.55c, any pixel that changed its class is highlighted—in this dynamic landscape, that includes nearly 50% of the area. Figure 7.55d focuses on one particular type of change, from shrubland to agriculture (approximately 15% of the area experienced this change).

Another approach to change detection using spectral pattern recognition is simply the *classification of multitemporal data sets*. The spectral bands for two (or more) dates of imagery are “stacked” together to make a new multitemporal data set; for example, two dates of four-band QuickBird multispectral imagery could be stacked together in a new eight-band data set. Supervised or unsupervised classification is then used to classify the combined image into temporal-categorical “classes”; some of these classes will be stable (i.e., the same class on both dates), while other classes will represent individual types of change. The success of such efforts depends upon the extent to which “change classes” are significantly different spectrally from the “nonchange” classes. Also, the dimensionality and complexity of the classification can be quite great, and if all bands from each date are used, there may be substantial redundancy in their information content.

Principal components analysis is sometimes used to analyze multirate image composites for change detection purposes. In this approach, two (or more) images are registered to form a new multiband image containing various bands from each date. Several of the uncorrelated principal components computed from the combined data set can often be related to areas of change. One disadvantage to this process is that it is often difficult to interpret and identify the specific nature of the changes involved. Plate 31 illustrates the application of multirate principal components analysis to the process of assessing tornado

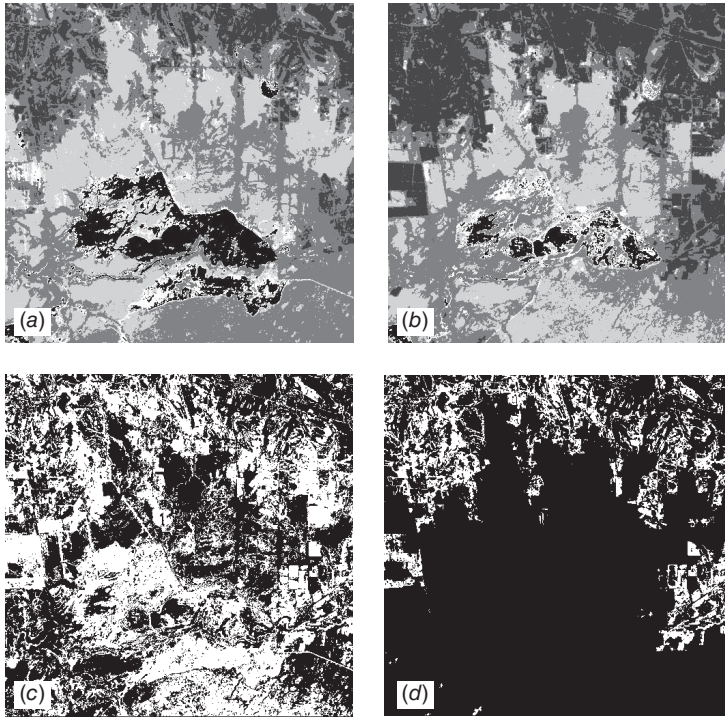


Figure 7.55 Postclassification change detection: (a) classification on date 1; (b) classification on date 2; (c) all change pixels highlighted in white; (d) only pixels that changed from class “shrubland” to class “agriculture.” (Author-prepared figure.)

damage from “before” and “after” images of the tornado’s path of destruction. The “before” image shown in (a), is a Landsat-7 ETM+ composite of bands, 1, 2, and 5 shown as blue, green, and red, respectively. The “after” image, shown in (b), was acquired 32 days later than the image shown in (a), on the day immediately following the tornado. While the damage path from the tornado is fairly discernable in (b), it is most distinct in the principal component image shown in (c). This image depicts the second principal component image computed from the six-band composite formed by registering bands, 1, 2, and 5 from both the “before” and “after” images.

Temporal image differencing is yet another common approach to change detection. In the image differencing procedure, DNs from one date are simply subtracted from those of the other. The difference in areas of no change will be very small (approaching zero), and areas of change will manifest larger negative or positive values. If 8-bit images are used, the possible range of values for the difference image is -255 to $+255$, so normally a constant (e.g., 255) is added to each difference image value for display purposes. Figure 7.56c illustrates the

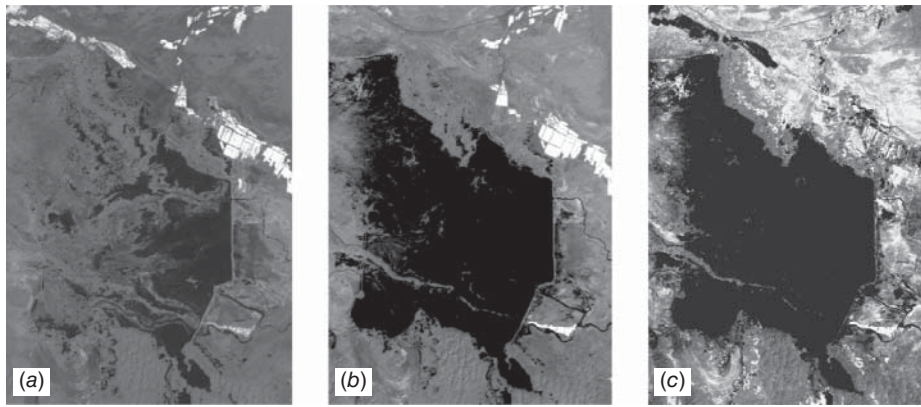


Figure 7.56 Temporal image differencing: (a) near-IR band from date 1; (b) near-IR band from date 2; (c) difference of (b) minus (a). (Author-prepared figure.)

result of temporal image differencing, with the two original (near-IR) images shown in (a) and (b). In this rendition, areas with no change have a mid-gray tone; areas whose near-IR reflectance increased are shown in lighter tones; and areas where the near-IR reflectance decreased are in darker tones. (Positive and negative changes are scaled symmetrically around the middle of the range.)

Temporal image ratioing involves computing the ratio of the data from two dates of imaging. Ratios for areas of no change tend toward 1 and areas of change will have higher or lower ratio values. Again, the ratioed data are normally scaled for display purposes (Section 7.6). One of the advantages to the ratioing technique is that it tends to normalize the data for changes in such extraneous factors as sun angle and shadows.

Whether image differencing or ratioing is employed, the analyst must find a meaningful “change–no change threshold” within the data. This can be done by compiling a histogram for the differenced or ratioed image data and noting that the change areas will reside within the tails of the distribution. A variance from the mean can then be chosen and tested empirically to determine if it represents a reasonable threshold. The threshold can also be varied interactively in most image analysis systems so the analyst can obtain immediate visual feedback on the suitability of a given threshold.

In lieu of using raw DNs to prepare temporal difference or ratio images, it is often desirable to correct for illumination and atmospheric effects and to transform the image data into physically meaningful quantities such as radiances or reflectances (Section 7.2). Also, the images may be prepared using spatial filtering or transformations such as principal components or vegetation components. Likewise, linear regression procedures may be used to compare the two dates of imagery. In this approach a linear regression model is applied to predict data values for date 2 based on those of date 1. Again, the analyst must set a threshold for detecting meaningful change in land cover between the dates of imaging.

Change vector analysis is a change detection procedure that is a conceptual extension of image differencing. Figure 7.57 illustrates the basis for this approach in two dimensions. Two spectral variables (e.g., data from two bands, two vegetation components) are plotted at dates 1 and 2 for a given pixel. The vector connecting these two data sets describes both the magnitude and direction of spectral change between dates. A threshold on the magnitude can be established as the basis for determining areas of change, and the direction of the spectral change vector often relates to the type of change. For example, Figure 7.57b illustrates the differing directions of the spectral change vector for vegetated areas that have been recently cleared versus those that have experienced regrowth between images.

One of the more efficient approaches to delineating change in multiband imagery is use of a *change-versus-no-change binary mask to guide multiband classification*. This method begins with a traditional classification of one image as a reference (time 1). Then, one of the spectral bands from this date is registered to the same band in a second date (time 2). This two-band data set is then analyzed using one of the earlier described algebraic operations (e.g., image differencing or ratioing). A threshold is then set to separate areas that have changed between dates from those that have not. This forms the basis for creating a binary mask of change-versus-no-change areas. This mask is then applied to the multiband image acquired at time 2 and only the areas of change are then classified for time 2. A traditional postclassification comparison is then performed in the areas known to have changed between dates.

This discussion has only provided a brief overview of the diversity of methods that have been developed for detection of changes between pairs of remotely

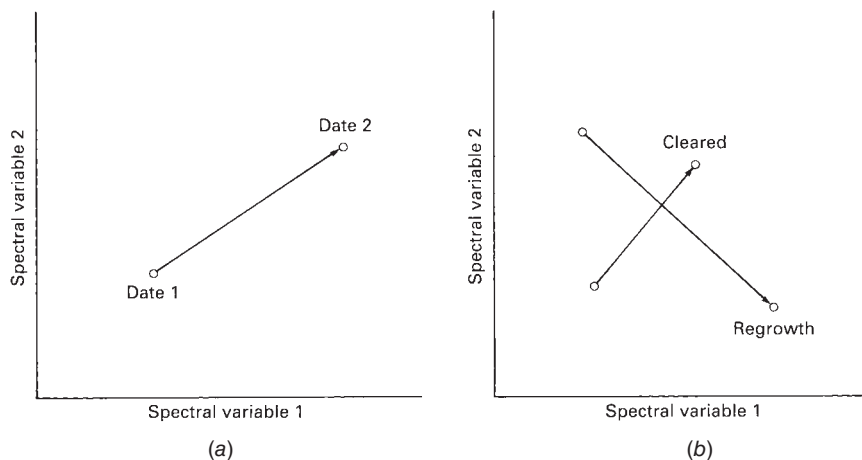


Figure 7.57 Spectral change vector analysis: (a) spectral change vector observed for a single land cover type; (b) length and direction of spectral change vectors for hypothetical "cleared" and "regrowth" areas.

sensed images. The automation of change detection, and of methods to interpret the significance of changes, is an area of great research interest. Given the ever-increasing constellation of airborne and spaceborne remote sensing platforms (with the role of UAVs expanding particularly rapidly), the volume of data being collected is immense, and automated methods to detect and characterize changes are correspondingly more important than ever.

7.19 IMAGE TIME SERIES ANALYSIS

The change detection methods discussed in the previous section are most commonly applied to small numbers of images from distinct times. The availability of long-term, systematically collected global data from sensors such as AVHRR, MODIS, and others provides the opportunity for obtaining much denser and richer temporal information. For any given location on the Earth, images from these “global monitoring systems” can be used to produce a time series of imagery and derived products (e.g., vegetation indices) at daily to weekly to monthly intervals, over a period of many years. These image time series are the basis for many studies of seasonal to decadal variations in agricultural, ecological, hydrological, and other regional-scale systems.

Figures 7.58 and 7.59 illustrate the concepts involved in a dense time series of imagery. The image in Figure 7.58 is a single date of NDVI (vegetation index) data derived from MODIS imagery in June 2011 for a site in China’s far western Xinjiang region. Although it contains the usual two-dimensional information about the distribution of NDVI values on the landscape, this image also can be seen as just one link in a long chain of 544 images extending forward and backward in time. (Additional NDVI images of this same area from Landsat imagery at finer scale were shown earlier in Figure 7.22.)

For any given pixel or group of pixels in this series of 544 dates of imagery, a “temporal signature” can be extracted, showing the seasonal to decadal evolution of NDVI within the pixel(s). Figure 7.59 shows four such series, for the points labeled *a* through *d* on the image in Figure 7.58. Each of these points represents a unique landscape feature with its own cyclical patterns and long-term trends in the vegetation index data. Location *a* represents existing farm fields planted with cotton (a crop that requires heavy irrigation in this arid region). The seasonal cycle in NDVI is distinct, and obviously derives from the recurring annual cycle of planting, growth, harvest, and senescence in the cotton fields. Location *b* began the decade as undeveloped, sparsely vegetated land with scattered desert shrubs growing on sandy soil. In 2006–2007, new farm fields were constructed in this area, and from 2008 onward, this site shows a pattern that is clearly similar to the existing agricultural fields at *a*. The area around *c* also was undeveloped land at the beginning of the decade, with shrubs (tamarisk) and scattered patches of *Populus euphratica* along current and former stream channels. In mid-2000, due to abnormally high flow in the main rivers in this area, there was extensive

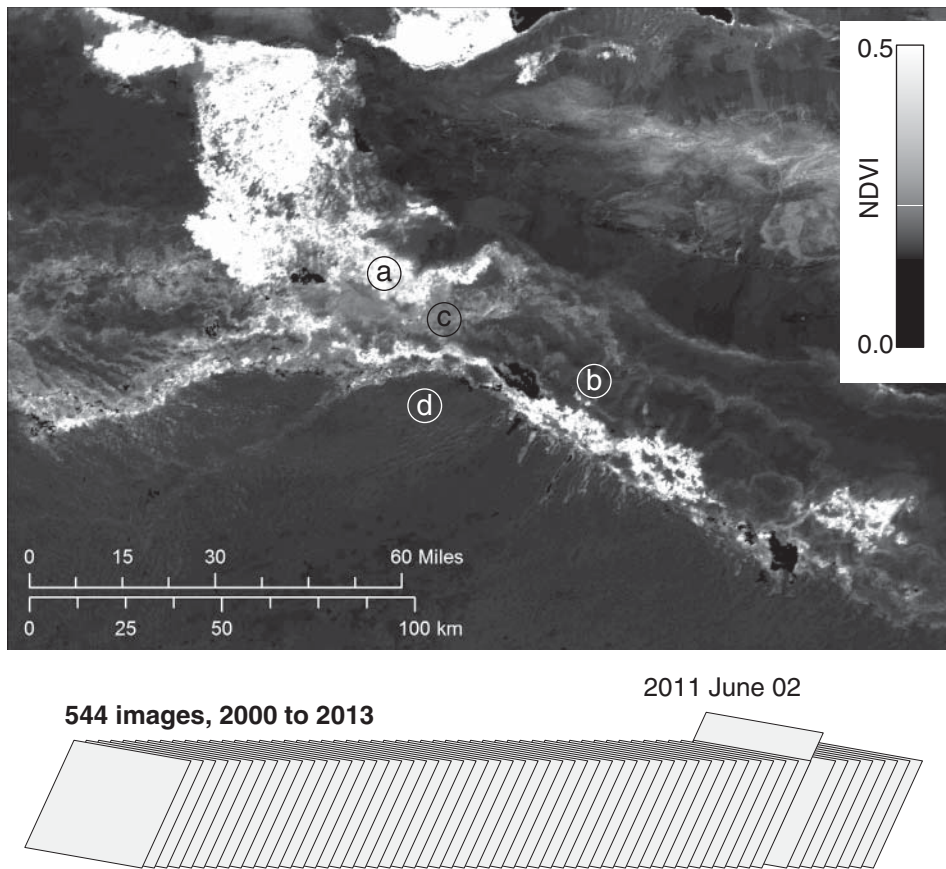


Figure 7.58 One typical date of MODIS NDVI imagery extracted from a time series of 544 images for the Tarim River area in Xinjiang, China, from February 2000 through February 2013. Labeled points (a–d) are sources for time-series graphs shown in Figure 7.59. (Author-prepared figure.)

flooding around site *c*. This addition of water produced a sudden increase in vegetation greenness over the next few years, manifested in the time series as an increase in the annual average NDVI, annual peak NDVI, and seasonal amplitude of NDVI. However, with no further recurrence of wide-scale flooding during the subsequent decade, the NDVI signal gradually declined back toward its previous state. For comparison, the temporal signature at *d*—basically, a flat line—is representative of unvegetated areas (in this case, large sand dunes in the Taklimakan Desert south of the river).

Plate 18 shows another analysis of an NDVI time series, in this case for the annual cycle of green-up and senescence across the United States. The analysis of these series of images representing spatial-temporal variation in vegetation indices (or other remotely sensed indices) is sometimes referred to as *landscape*

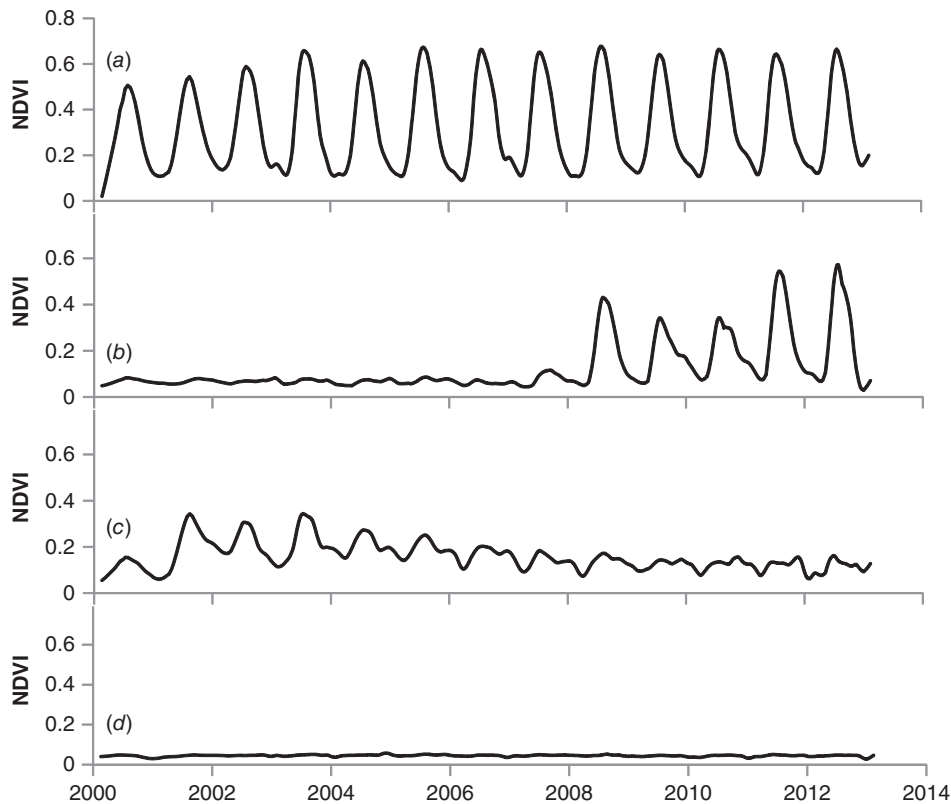


Figure 7.59 Time-series plots of MODIS NDVI for points shown in Figure 7.58: (a) established cotton farms; (b) new cotton farms; (c) tamarisk shrubland following an episode of flooding; (d) sand dunes.

phenology. It is increasingly being used as the basis for research on regional-scale linkages between climate and ecosystems (White et al., 2009), as well as for tasks as diverse as agricultural yield forecasting, hydrologic monitoring, and studies of changes in snow and ice cover.

Plate 32 shows several representations of a 12-year (2000 to 2011) time series of MODIS enhanced vegetation index (EVI; see Section 7.6) data for the region around Tarangire National Park in northern Tanzania. In (a), the long-term mean EVI value is shown for each pixel, with greener hues representing more dense vegetation. This region consists of savannas at lower elevations, with forests growing at middle elevations on isolated volcanic peaks such as Mt Kilimanjaro (upper right). Using the full time-series of images, a linear decadal trend was calculated for every pixel (b). The greenest areas in Plate 32b represent increasing trends of over 0.1 EVI units per decade, while the reddest areas represent correspondingly large negative trends. Finally, part (c) shows the amplitude of the normal seasonal cycle in each pixel, with blue areas having very little peak-to-trough

variation in EVI each year, while orange or brown areas manifest a much stronger annual signal in growth and senescence at the landscape scale. (In all three parts of this plate, a subtle topographic shading effect has been added, to enhance the interpretability of the images.) Researchers are using these data to understand the landscape dynamics of this region and the factors influencing the movements of large migratory ungulates (e.g., wildebeest) over time.

From looking at the time series graphs (or *multitemporal profiles*) shown in Figure 7.59a–c, it can be seen that both agricultural and natural systems have their distinctive seasonal patterns of greenness, often with regularly timed cycles that repeat each year. Scientists use these profiles for classification (of agricultural crops and natural ecosystems) as well as for estimating factors such as water demand, productivity, and other biophysical quantities at the landscape scale. These analyses are based on physical modeling of the time behavior of each crop or ecological community's spectral response pattern. It has been found that the time behavior of the greenness of annual crops is sigmoidal (Figure 7.60), whereas the greenness of the soils (G_0) in a given region is nearly constant. Thus, the greenness at any time t can be modeled in terms of the peak greenness G_m , the time of peak greenness t_p , and the width σ of the profile between its two

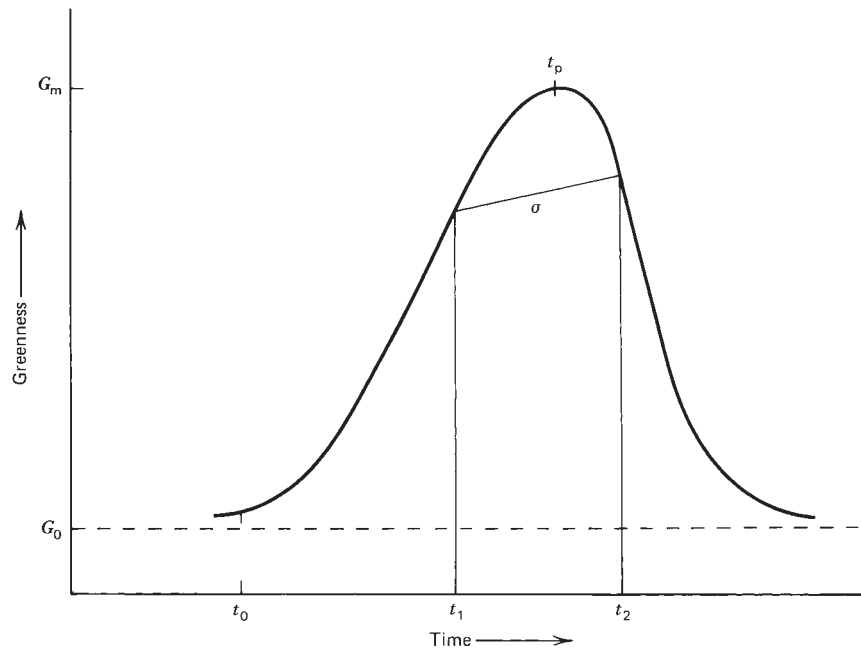


Figure 7.60 Temporal profile model for greenness. Key parameters include spectral emergence date (t_0), time (t_p) of peak greenness (G_m), and width of the profile (σ). (Adapted from Bauer, 1985, after Badhwar, 1985.)

inflection points. (The inflection points, t_1 and t_2 , are related to the rates of change in greenness early in the growing season and at the onset of senescence.) The features G_m , t_p , and σ account for more than 95% of the information in the original data and can therefore be used for classification and modeling, instead of the original spectral response patterns. These three features are important because they not only reduce the dimensionality of the original data but also provide variables directly relatable to agrophysical parameters.

7.20 DATA FUSION AND GIS INTEGRATION

Many applications of digital image processing are enhanced through the merger or *fusion* of multiple data sets covering the same geographical area. These data sets can be of a virtually unlimited variety of forms. We have already discussed, or alluded to, several applications of data fusion in remote sensing. For example, one frequently applied form of data merger is the combining of multiresolution data acquired by the same sensor, such as the “pan-sharpening” approach shown in Plate 16 that merges IKONOS 1-m panchromatic and 4-m multispectral data. In another example, in Figure 7.54, an artificial neural network was used to combine multispectral imagery with radar-derived texture and topographic slope and aspect (from a DEM) for image classification.

Often, remotely sensed data and their derived products are integrated with other spatial data in a GIS. In Plate 2, we illustrated the merger of automated land cover classification data with soil erodibility and slope information in a GIS environment in order to assist in the process of soil erosion potential mapping. Over time, the boundaries between digital image processing and GIS operations have become blurred, and fully integrated spatial analysis systems have become the norm. In the following discussion, we begin with the simplest case—merging multiple data sets from the same type of sensor—then proceed to discuss multi-sensor image fusion, and finally the integration of remotely sensed data with other types of data.

Multitemporal Data Fusion

In Sections 7.18 and 7.19, we examined the use of multitemporal data for detecting changes on the landscape and for studying seasonal to interannual variability in landscape properties such as vegetation cover. In other cases, multitemporal images are “fused” together to create a product useful for visual interpretation. For example, agricultural crop interpretation is often facilitated through merger of images taken early and late in the growing season. In early season images from mid-latitude temperate agricultural regions, bare soils often appear that later will be planted in such crops as corn or soybeans. At the same time, early season images might show perennial alfalfa or winter wheat in an advanced state of